

Leakage-Aware Transaction Risk Classification with Jev and Tree Ensembles

Abstract. Post-event fields can make transaction-risk classification appear easier than it is at screening time. We evaluate a hosted Jev decision service, tuned Random Forest and XGBoost classifiers, and a prevalence-only reference on one simulated accounting-transaction dataset. All receive the same 12 leakage-screened features for 2,000 held-out records; the trees are trained on 8,000 labelled records, whereas Jev receives no dataset-specific examples. Three excluded post-event fields perfectly encode the outcome. With 14.5% positive test labels, ROC-AUC is 0.503 for Random Forest, 0.515 for XGBoost, and 0.492 for Jev. Jev reaches 0.700 recall by flagging 71.9% of cases, with 0.141 precision. The constant-prevalence reference has a lower Brier score (0.124) than all three substantive models (0.243–0.459). Filtering by Jev’s reported confidence reduces coverage without improving accepted-case accuracy at the tested cutoffs. This leakage-aware evaluation finds little usable discrimination in the permitted features; neither variable probability scores nor numerically high confidence establish reliable individual risk decisions on this dataset. The unequal supervision and simulated labels limit comparisons beyond this setting.

Keywords: Tabular classification · Financial risk · Data leakage · Probability reliability · Selective prediction

1 Introduction

Transaction risk screening is a structured decision problem: an automated system must use information available when a transaction is assessed, not fields recorded after an incident is classified. This matters when datasets mix transaction attributes with incident descriptions. An evaluation using those descriptions answers an operationally invalid question, even if its scores are high. Cross-domain analyses document how illegitimate predictors and evaluation leakage distort model comparisons [15, 1]. Recent financial-risk preprints likewise call attention to protocol choice, although they study different tasks and datasets [17, 23].

Random Forest and XGBoost are established supervised choices for tabular classification [2, 3]. Large tabular benchmarks support strong tree baselines but show that results vary with datasets and tuning [8, 16]. A tabular foundation model such as TabPFN instead uses labelled in-context examples [14]; zero-shot language-model studies investigate decisions without these examples and report mixed results [9, 7].

Jev is TypeSafe’s hosted System One model. According to its documentation, it evaluates an input *state* against typed questions and returns constrained

answers rather than generated prose [22]. For a **Choice** question, the caller supplies instructions and descriptions of the possible labels; the response contains the selected label, a probability for each option, and confidence derived from the spread of those probabilities [20, 21]. Here the state is one transaction and the options are **risk_incident** and **no_risk_incident**. These documented interface properties do not establish Jev’s undisclosed architecture, training data, or calibration on our labels.

Existing work considers leakage-safe evaluation, supervised tabular learning, zero-shot tabular decisions, probability calibration, and prediction with a reject option, often separately. It gives little independent evidence about a hosted decision-oriented model and supervised tree baselines under the same leakage-screened financial-risk protocol. In particular, whether a model’s confidence usefully selects automatic decisions when admissible inputs carry little ranking signal remains unresolved for this setting.

Classification accuracy alone is difficult to interpret with a minority risk class: predicting no incident for every case can produce high accuracy while missing every incident. Precision–recall analysis addresses this imbalance [5]. Ranking also does not establish that predicted probabilities match event frequencies [10, 12]. Finally, selective classification requires checking accuracy at the achieved coverage, not merely thresholding a high confidence number [13, 19].

This study asks four questions on one common held-out test set: (RQ1) how do the systems classify risk incidents; (RQ2) how reliable are their positive-class probabilities; (RQ3) does Jev confidence-based deferral improve accepted-case performance; and (RQ4) what latency trade-off accompanies local and hosted inference? The contribution is an empirical, leakage-aware comparison across these dimensions, including a prevalence-only reference classifier. We report weak discrimination without selecting an outcome threshold or prompt after inspecting test labels. We do not estimate a before-and-after performance change from removing leakage.

2 Method

2.1 Dataset and Feature Availability

We use version 1 of the Financial Transaction and Risk Management Dataset published by Ziya on Kaggle under CC0 [24]. Its description says it was designed to *simulate* accounting transactions and risk events. Accordingly, the dataset is a simulation, not evidence of performance in a financial institution. The supplied CSV contains 10,000 rows and 18 columns, including a binary **Risk_Incident** label (1,448 positives; 14.48%). The audit finds no duplicate records, duplicate transaction IDs, or empty CSV values. Transaction dates range from January 2024 to January 2025. The source dataset description does not specify an independently validated incident-label generation rule.

Our feature-availability rule retains a predictor only if its value could plausibly be known before or at transaction screening; fields describing a subsequently

Table 1. Dataset and leakage-aware evaluation protocol. Test counts come from the frozen prediction files.

Dataset / target	Simulated accounting transactions / Risk.Incident
Rows / positive labels	10,000 / 1,448 (14.48%)
Training / test rows	8,000 / 2,000 (seed 42, stratified)
Positive / negative test labels	290 / 1,710
Model inputs	4 categorical + 8 numeric/derived fields
Target-encoding fields excluded	Risk.Type, Incident.Severity, Error.Code

identified incident or its severity are post-outcome inputs. The audit shows that **Risk.Type**, **Incident.Severity**, and **Error.Code** each perfectly separate the binary classes in this file: their non-incident placeholder and incident values encode the target. None enters a classifier or Jev state. We also exclude transaction/account/user identifiers and **Counterparty** to avoid memorising high-cardinality identities; **Currency** is constant. The original **Date** becomes year, month, day of week, and day of year. The remaining 12 features comprise transaction type, category, payment method, IP region, amount, system latency, login frequency, failed attempts, and these four calendar components. Jev and the supervised baselines use the same feature names and underlying values, although their representations differ. Availability of system latency and behavioural fields at the instant of screening is assumed, not independently established by the dataset publisher. The parser preserves the valid literal **NA** IP-region code and does not misinterpret the **None** placeholder in excluded columns as missing data.

2.2 Split and Model Configurations

A single stratified 80/20 split with random seed 42 fixes the held-out test IDs for every method. This permits paired per-record comparisons without repeating hosted Jev requests across splits; split variability is therefore outside the scope of our intervals. Supervised preprocessing is fitted inside each training pipeline: median imputation for numeric fields, most-frequent imputation and one-hot encoding for categorical fields, with unseen categories ignored. On the training portion only, five-fold shuffled stratified cross-validation selects hyperparameters by average precision (AP). Random Forest uses balanced class weights; its selected configuration has 200 trees, maximum depth 10, and minimum split size 5. XGBoost uses **scale_pos_weight** equal to the training negative/positive ratio, 0.8 row and column subsampling, histogram tree construction, and a selected 200 trees, depth 3, and learning rate 0.03. Its evaluation metric during fitting is log loss. The chosen parameter values, package versions, and split are recorded in the experiment metadata. A **DummyClassifier** predicting according to training prevalence supplies a non-discriminating reference. Supervised hard predictions use a fixed positive-probability threshold strictly greater than 0.5; this threshold was not tuned on the test set.

Jev is evaluated as a hosted, task-untrained decision service. For each held-out row, the same 12 feature values are serialised as a named state and sub-

mitted with a binary **Choice** question: “Classify the transaction in the supplied state. Choose `risk_incident` when the available transaction information indicates that this transaction should be flagged as a risk incident; otherwise choose `no_risk_incident`. Use only the supplied state.” The two choice criteria restate whether the transaction should or should not be flagged. No training labels, exemplars, or excluded incident fields are supplied. A schematic state with all twelve feature names is:

```
{ "Transaction_Type": "Transfer", "Category": "Other",
  "Payment_Method": "Card", "IP_Region": "NA",
  "Amount": 100.0, "System_Latency": 120.0,
  "Login_Frequency": 3, "Failed_Attempts": 0,
  "date_year": 2024, "date_month": 7,
  "date_dayofweek": 0, "date_dayofyear": 183 }
```

It is paired with the quoted **Choice** instructions and criteria `risk_incident` (should flag) and `no_risk_incident` (should not flag). The values are illustrative, not an observed dataset row.

The client requested `jev-latest`; the pilot and saved responses identify the returned model as `jev-1.13.0` (pilot recorded 25 September 2026). The question version is `binary-risk-choice-v1`. The chosen option determines Jev’s hard prediction. Its `probabilities["risk_incident"]` supplies the positive-class probability. The returned `confidence` is computed from the option probabilities according to TypeSafe’s documentation [21]; we use it only for deferral analysis, not as an independent estimate or a calibrated probability of correctness. Unlike Random Forest and XGBoost, Jev did not fit on this dataset’s training labels; consequently this is a comparison of decision settings, not a controlled comparison of learning algorithms.

2.3 Measurements and Uncertainty

We report accuracy, positive-class precision, recall, F1, ROC-AUC, AP, and Brier score. AP summarises the precision–recall ranking; its non-informative reference level is approximately the positive prevalence. For labels $y_i \in \{0, 1\}$ and predicted positive probabilities p_i , the Brier score is $N^{-1} \sum_i (p_i - y_i)^2$ (lower is better). We also inspect ten-quantile-bin reliability plots. Brier score is a proper scoring rule that mixes calibration and refinement; a low score *alone* does not prove useful discrimination or good calibration [4].

For Jev, we retain a prediction when its reported confidence is at least $t \in \{0.50, 0.60, 0.70, 0.80, 0.90, 0.95\}$. Coverage is the retained fraction of all 2,000 test records; performance is computed *only among retained records*, and the others count as prospective human-review cases. These thresholds describe test-set behaviour rather than an independently validated deployment policy. For uncertainty, 1,000 seeded bootstrap resamples of shared held-out IDs provide percentile 95% intervals for individual metrics and paired model differences; they do not measure variation from a new train/test split or new Jev request. Local model latency is wall-clock time for one row’s pipeline `predict_proba`;

Jev latency is the hosted API round trip, including network effects. Both are summarised by mean, median, and 95th percentile, but are not measurements of identical compute boundaries.

3 Results

3.1 Classification and Probability Scores

Table 2 reports the full 2,000-row held-out evaluation. All three non-dummy ROC-AUC values are close to 0.5 and their 95% bootstrap intervals include 0.5: Random Forest 0.503 [0.466, 0.541], XGBoost 0.515 [0.479, 0.555], and Jev 0.492 [0.456, 0.533]. AP ranges from 0.151 to 0.164, near the 0.145 test prevalence. The paired XGBoost-minus-Jev ROC-AUC interval is [-0.027, 0.071]; the paired F1 interval is [-0.046, 0.012]. Neither supports a clear difference on these two metrics at this sample size. This is not evidence of equivalence.

Table 2. Held-out classification results ($N = 2,000$). AP is average precision, not trapezoidal PR area; BS is Brier score (lower is better). The dummy predicts no incident for all rows.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	AP	BS↓
Prevalence dummy	.855	.000	.000	.000	.500	.145	.124
Random Forest	.626	.152	.345	.211	.503	.152	.243
XGBoost	.542	.145	.441	.218	.515	.164	.244
Jev	.339	.141	.700	.235	.492	.151	.459

The dummy’s 0.855 accuracy results from predicting the majority class and missing all 290 incidents. Jev predicts risk for 1,438 of 2,000 transactions (71.9%): 203 true positives, 1,235 false positives, 87 false negatives, and 475 true negatives. Random Forest yields 100 true positives and 559 false positives; XGBoost yields 128 and 754, respectively. Jev’s higher recall and F1 at the fixed operating point therefore coincide with many extra alerts, while its ROC ranking remains near chance. The precision–recall curves in Fig. 1 show how precision varies with recall.

On probabilistic scoring, Jev’s Brier score is 0.459 versus 0.243 for Random Forest, 0.244 for XGBoost, and 0.124 for the constant-prevalence dummy. The paired Jev-minus-XGBoost Brier difference is 0.215 with a 95% interval of [0.201, 0.230]. The dummy’s lower score is central to interpreting this negative result: on these records, varying the estimated probability by transaction worsens squared-error prediction relative to using training prevalence alone. This does not make the dummy a useful incident detector; it has no ranking or individual case-selection ability. Reliability curves (Fig. 2) provide a complementary descriptive view.

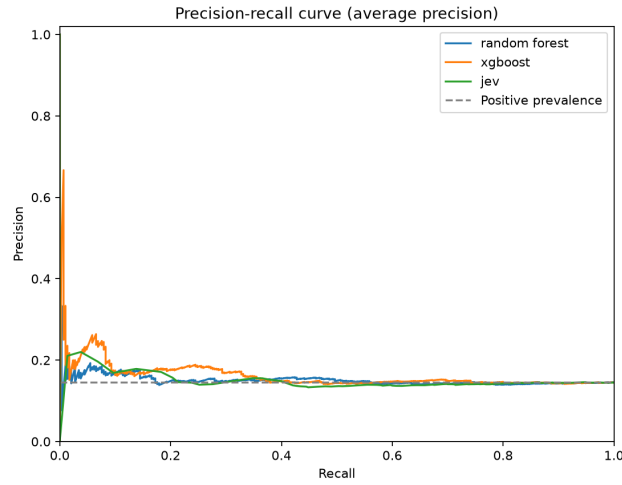


Fig. 1. Precision–recall curves for the three substantive systems on the held-out set. The dashed reference is positive prevalence; the non-discriminating dummy is reported in Table 2 rather than drawn as a curve. AP values are reported in the table. Generated from saved predictions.

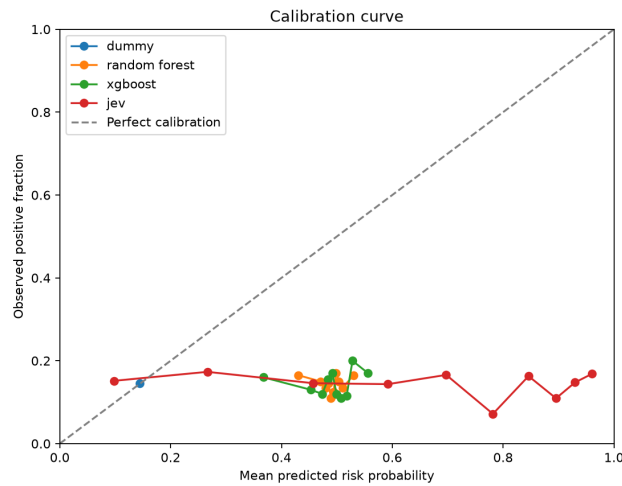


Fig. 2. Positive-class reliability curves with up to ten quantile bins per model. For the three variable-score models, bins contain approximately 200 records each unless tied scores collapse a boundary; the constant-score dummy contributes one point for all 2,000 cases. The diagonal represents perfect calibration. Points occur at different mean scores across models, and empty bins are omitted.

3.2 Confidence-Based Deferral and Latency

Table 3 shows that increasing the Jev confidence cutoff reduces accepted-case coverage from 63.7% at 0.50 to 1.65% at 0.95. At every listed cutoff, accepted-case accuracy (0.272–0.310) is below its unfiltered 0.339 accuracy; these thresholds therefore do not identify a more accurate automatic-decision subset. F1 increases at some cutoffs, but class composition changes among accepted records and the number sent for review grows sharply. At 0.90, only 194 cases are retained and 1,806 are deferred; at 0.95, only 33 are retained and accuracy is 0.273. These are descriptive comparisons on selected subsets, not estimates of how a staffed review workflow would perform.

Table 3. Jev selective prediction. Metrics are conditional on accepted cases; review count includes rejected cases. Unfiltered accuracy and F1 are .339 and .235.

Confidence t	Coverage	Accepted	Review	Accuracy	Precision	Recall	F1
.50	.637	1,274	726	.302	.134	.734	.226
.60	.548	1,096	904	.308	.142	.753	.239
.70	.435	869	1,131	.310	.157	.779	.261
.80	.286	571	1,429	.280	.141	.813	.240
.90	.097	194	1,806	.299	.177	.824	.292
.95	.017	33	1,967	.273	.192	.625	.294

The per-record Random Forest median is 14.8 ms (p95 14.9 ms), versus 1.5 ms (p95 1.7 ms) for XGBoost. Jev’s hosted round-trip median is 264.0 ms (p95 311.6 ms). The respective means are 14.6, 1.5, and 270.9 ms. These values reflect this run and execution environment; Jev’s measurement includes service and network time and is not a pure model-compute benchmark.

4 Discussion

4.1 Interpreting the Research Questions

RQ1: classification. Near-chance ROC-AUC and AP close to prevalence suggest weak ranking signal in the permitted features for this particular split. Jev’s high recall follows from flagging most transactions, not from convincing discrimination. The dummy’s majority-class accuracy shows why an accuracy-only model ranking would misrepresent incident-detection utility. Benchmarks favouring trees on some tabular datasets do not imply that any classifier can recover information missing from admissible features [8, 16]. Recent fraud preprints report stronger discrimination with different labels, inputs, and protocols [11, 6]; their scores are not comparable with these simulated labels. Here, excluding fields that perfectly encode the label establishes a defensible screening task, but we did not run a leakage-inclusive performance benchmark from which to estimate a performance drop [15].

RQ2: probability reliability. The Brier scores and reliability plot show that the probabilities from these runs should not be read as dependable individual risk likelihoods. A constant predictor using training prevalence beats every

evaluated model on Brier score despite detecting no incidents. Thus variation in the three systems’ scores adds no squared-error benefit over the base rate on this test set. This does not imply that the dummy’s constant score is an adequate screening policy: it cannot rank cases. Calibration and refinement must be considered alongside discrimination [4]. Tabular recalibration methods can improve probability estimates on other datasets [12], but evaluating one here would require separate calibration data and would not establish that the available features identify individual incidents.

RQ3: selective decisions. Jev’s confidence is derived from the concentration of its reported option probabilities [21]; a numerically high value is not evidence that the selected decision matches this dataset’s label. High confidence did not identify a subset with higher accuracy at our cutoffs. Although some accepted-subset F1 values rise, these are conditional on changing class composition and require deferring most cases at high thresholds. Reject-option research distinguishes predictive quality on retained cases from the fraction rejected [13]; a recent high-stakes preprint also stresses review capacity and costs [19]. A separate preprint auditing a commercial System-1 model found sensitivity to answer-option presentation [18]. Neither study validates the present Jev scores or a staffed review policy.

RQ4: operation. Jev’s observed round trip is slower than local XGBoost inference; Random Forest is also slower per row than XGBoost in this implementation. Different timing boundaries, hardware, API load, network path, and batch sizes limit the generality of these numbers. No cost estimate or production-service guarantee follows from this run.

4.2 Validity and Scope

First, the publisher describes the dataset as a simulation, but does not publish a verified real-world outcome mechanism. The perfect target encoding in three incident fields demonstrates why feature availability must be specified. It does *not* establish that the remaining 12 fields carry a usable pre-event signal, nor prove that labels were generated independently of them. Thus we cannot attribute the weak discrimination to any one model architecture. Second, one random stratified split of one dataset does not estimate temporal generalisation, robustness to changed prevalence, or variability over data splits. Bootstrap intervals quantify uncertainty conditional on this frozen test set and fixed predictions.

Third, the methods receive unequal supervision: the trees see 8,000 labelled records, while Jev sees one record at a time with no examples. No Jev model card or training-data audit was available to establish what the hosted system may have encountered during pretraining. Its alias can change; the returned version and prompt version are recorded for this run. Only one fixed question, set of option descriptions, and feature serialisation were evaluated; sensitivity to wording or feature and option order remains unknown. The question asks whether a row *should* be flagged without an externally validated rule linking its observable fields to the synthetic incident label. More detailed incident criteria cannot be reconstructed from the excluded post-event fields without leakage. Fourth,

the Jev hard choice need not coincide with thresholding its reported positive-class probability at 0.5; we respect the returned decision rather than retrofitting a test-set threshold. Selected-case metrics at the six cutoffs have no separate selection dataset and should not be presented as a deployable deferral policy. Human reviewers, error costs, and downstream actions were not measured.

Finally, the tabular foundation model and LLM studies cited above involve different architectures, context regimes, and datasets [14, 9, 7]. Their mixed evidence motivates careful evaluation, not a transfer of their performance claims to Jev. Likewise, protocol-sensitive risk benchmarking elsewhere [17, 23] reinforces the need to keep leakage control and dataset provenance explicit when interpreting this negative finding.

5 Conclusion

On a shared, leakage-aware evaluation of simulated accounting transactions, Jev, Random Forest, and XGBoost all yield ROC-AUC values near chance after target-encoding fields are removed. Jev flags many more cases and attains higher recall, but at low precision and a substantially worse Brier score than the local models; its reported confidence does not improve accepted-case accuracy at the tested cutoffs. These observations bound what can be claimed from the experiment: neither the hosted decision service nor the supervised baselines demonstrate reliable individual incident screening on the permitted features. A useful next study would test independently justified pre-event risk criteria and additional datasets with documented label generation, while fixing prompts, thresholds, and review budgets before examining test labels.

AI Usage Declaration

ChatGPT and OpenCode were used to assist with experimental code, evidence organisation, an initial manuscript draft, bibliography preparation, and language revision. The authors reviewed the code and take responsibility for verifying the data, results, sources, figures, interpretations, and final wording. Generative AI did not generate the research data or experimental figures.

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